

THE UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY



Kiểm thử phần mềm - 22KTPM3

SCENARIO TESTING - HW3

Thành Viên

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Chapter 1

Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	- SignUp	Done
		- Checkout	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
		- User Management	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

Chapter 2

Scenario Definition

2.1 Feature: Catalog

2.1.1 Basic Flow: View and Interact with Catalog

- **Description:** The user views the catalog, applies sorting, filtering, price range, search, and pagination to explore products.
- **Preconditions:** Catalog page is accessible, and the system has preloaded products (e.g., `total_pages = 3`).
- **Steps:**
 1. User navigates to the Catalog page (Home page) (<http://localhost:4200>).
 2. User selects options from the Sort dropdown.
 3. User enters the `min_price` field and the `max_price` field, then applies the price range filter.
 4. User checks checkboxes under Filters and applies the filter.
 5. User enters characters in the search field and submits the search.
 6. User verifies the product list is sorted, filtered, and matches the search term.
 7. User navigates to page 2 using the “Next” button.
 8. System displays successfully.
- **Expected Result:** Products are displayed in ascending order by price between `min_price` and `max_price`, filtered, matching the search, and pagination works correctly.

2.1.2 Alternate Flows

- **Alternate Flow 1: Empty Search with Filters**
 - Trigger: Step 5, user submits an empty search field after applying filters.

- Expected Result: Products are displayed based on the “Hammer” filter, ignoring the empty search, with no errors.
- **Alternate Flow 2: Pagination to Invalid Page**
 - Trigger: Step 7, user attempts to navigate to page 4.
 - Expected Result: Navigation stops at page 3, “Next” button is disabled, and an error or notice may be displayed.
- **Alternate Flow 3: Search with Maximum Length and Invalid Character**
 - Trigger: Step 5, user enters a 101-character string with a special character.
 - Expected Result: An error message is displayed (e.g., “Search input exceeds 100 characters or contains invalid characters”), and no results are shown.
- **Alternate Flow 4: Empty Sort Selection**
 - Trigger: Step 2, user leaves Sort as empty string (“”).
 - Expected Result: Products are filtered by “Hammer” and displayed in default order (e.g., database order), with no sorting applied.
- **Alternate Flow 5: Price Range with Min > Max**
 - Trigger: Step 3, user enters min_price greater than max_price.
 - Expected Result: An error message is displayed, or the filter is not applied, maintaining the previous state.
- **Alternate Flow 6: No Filters Applied**
 - Trigger: Step 4, user submits without selecting any filters.
 - Expected Result: All products are displayed without filtering.
- **Alternate Flow 7: Case-Insensitive Search Failure**
 - Trigger: Step 5, user enters an uppercase search term.
 - Expected Result: Products matching “hammer” are displayed regardless of case.

2.1.3 Scenario Table for Catalog

Scenario Name	Starting Flow	Alternate Flow
Scenario 1 - Successful Catalog View	Basic Flow	
Scenario 2 - Empty Search with Filters	Basic Flow	Alternate Flow 1
Scenario 3 - Pagination to Invalid Page	Basic Flow	Alternate Flow 2

Scenario Name	Starting Flow	Alternate Flow
Scenario 4 - Search with Maximum Length and Invalid Character	Basic Flow	Alternate Flow 3
Scenario 5 - Empty Sort Selection	Basic Flow	Alternate Flow 4
Scenario 6 - Price Range with Min > Max	Basic Flow	Alternate Flow 5
Scenario 7 - No Filters Applied	Basic Flow	Alternate Flow 6
Scenario 8 - Case-Insensitive Search Failure	Basic Flow	Alternate Flow 7

2.2 Feature: Categories

2.2.1 Basic Flow: View and Interact with Hand Tools or Power Tools Category

- **Description:** The user views the Hand Tools or Power Tools category, applies sorting, filtering, and pagination to explore products.
- **Preconditions:** Categories page is accessible (Hand Tools/Power Tools), and Hand Tools or Power Tools category has products (e.g., total_pages = 3).
- **Steps:**
 1. User navigates to the Categories and selects “Hand Tools” / “Power Tools”.
 2. User selects options from the Sort dropdown.
 3. User checks checkboxes under Filters and applies the filter.
 4. User verifies the product list is sorted, filtered.
 5. User navigates to page 2 using the “Next” button.
 6. System displays successfully.
- **Expected Result:** Products are displayed in filtered , sorted, and pagination works correctly.

2.2.2 Alternate Flows

- **Alternate Flow 1: Empty Sort Selection**
 - Trigger: Step 2, user leaves Sort as empty string (“”).
 - Expected Result: Products are filtered to show only items, displayed in default order (e.g., database order).

- **Alternate Flow 2: Pagination to Invalid Page**
 - Trigger: Step 7, user attempts to navigate to page 4.
 - Expected Result: Navigation stops at page 3, “Next” button is disabled, and an error or notice may be displayed.
- **Alternate Flow 3: Apply Filter with Maximum Selections**
 - Trigger: Step 3, user selects all available filters.
 - Expected Result: All products matching the selected categories and brand are displayed, sorted correctly.
- **Alternate Flow 4: No Filters Applied**
 - Trigger: Step 3, user submits without selecting any filters.
 - Expected Result: All products in the selected category (Hand Tools or Power Tools) are displayed without filtering.
- **Alternate Flow 5: Single Category Filter Failure**
 - Trigger: Step 3, user selects a non-existent category.
 - Expected Result: No products are displayed, or an error message is shown.
- **Alternate Flow 6: Empty Search with Filters**
 - Trigger: Step 3 (assuming search is available), user submits an empty search after filters.
 - Expected Result: Products are displayed based on the “Hammer” filter, ignoring the empty search.
- **Alternate Flow 7: Product Names Aligned to the Right**
 - Trigger: Step 4, user views the product list.
 - Expected Result: Product names are incorrectly aligned to the right instead of the expected left or center alignment.

2.2.3 Scenario Table for Categories

Scenario Name	Starting Flow	Alternate Flow
Scenario 1 - Successful Hand Tools View	Basic Flow	
Scenario 2 - Successful Power Tools View	Basic Flow	
Scenario 3 - Empty Sort Selection (Hand Tools)	Basic Flow	Alternate Flow 1
Scenario 4 - Empty Sort Selection (Power Tools)	Basic Flow	Alternate Flow 1

Scenario Name	Starting Flow	Alternate Flow
Scenario 5 - Pagination to Invalid Page (Hand Tools)	Basic Flow	Alternate Flow 2
Scenario 6 - Pagination to Invalid Page (Power Tools)	Basic Flow	Alternate Flow 2
Scenario 7 - Apply Filter with Maximum Selections (Hand Tools)	Basic Flow	Alternate Flow 3
Scenario 8 - Apply Filter with Maximum Selections (Power Tools)	Basic Flow	Alternate Flow 3
Scenario 9 - No Filters Applied (Hand Tools)	Basic Flow	Alternate Flow 4
Scenario 10 - No Filters Applied (Power Tools)	Basic Flow	Alternate Flow 4
Scenario 11 - Single Category Filter Failure (Hand Tools)	Basic Flow	Alternate Flow 5
Scenario 12 - Single Category Filter Failure (Power Tools)	Basic Flow	Alternate Flow 5
Scenario 13 - Empty Search with Filters (Hand Tools)	Basic Flow	Alternate Flow 6
Scenario 14 - Empty Search with Filters (Power Tools)	Basic Flow	Alternate Flow 6
Scenario 15 - Product Names Aligned to the Right (Hand Tools)	Basic Flow	Alternate Flow 7
Scenario 16 - Product Names Aligned to the Right (Power Tools)	Basic Flow	Alternate Flow 7

Chapter 3

Use of AI Tool

In the development of the scenario testing for the Catalog and Categories features, an AI tool was utilized to assist in the process. Below is a detailed report on the use of the AI tool, including its name, prompts used, validation and refinement process, and the distinction between AI-generated and manually created test cases.

3.1 Tool name

The AI tool used was ChatGPT, developed by OpenAI.

3.2 Prompt used

The following prompts were provided to ChatGPT to generate the initial content for the scenario testing:

- **Prompt to Generate Test Cases for Catalog:** “Design a detailed scenario-based testing plan for a Catalog feature of an e-commerce application, including Basic Flow, Alternate Flows, and a Scenario Table with fields: Scenario ID, Title, Preconditions, Inputs (Test Data), Test Steps, Expected Result, Actual Result, and Result.”
- **Prompt to Generate Test Cases for Categories:** “Create a scenario-based testing plan for a Categories feature with separate pages for Hand Tools and Power Tools, including Basic Flow, Alternate Flows, and a Scenario Table with fields: Scenario ID, Title, Preconditions, Inputs (Test Data), Test Steps, Expected Result, Actual Result, and Result. Include specific filters (e.g., Hammer, Grinder) and sort options, with total_pages = 3.”
- **Prompt to Generate Bug Report:** “Enhance the Catalog and Categories scenario tables with additional edge cases, such as invalid inputs, maximum filter

selections, and special characters in search, ensuring distinct Scenario IDs for Hand Tools and Power Tools. Based on the test case results, generate bug reports in the format: Bug ID, Summary, Steps to Reproduce, Expected Result, Actual Result, Screenshot, Priority, Severity.”

3.3 How you validated or refined the AI-Generated results

The scenario-based test cases generated by ChatGPT were carefully reviewed and validated through the following process:

1. Consistency Check:

- Verified that all AI-generated scenarios followed a uniform structure (e.g., Basic Flow, Alternate Flow, Scenario Table).
- Ensured that scenarios had clear preconditions, expected results, and logical flow of steps.

2. Cross-Referencing with Functional Requirements:

- Matched each scenario with the actual functional requirements of the Catalog and Categories features.
- Ensured that all user interactions (e.g., sorting, filtering, pagination, search) were covered.

3. Edge Case Validation:

- Reviewed AI-generated edge cases (like invalid page numbers or input length violations) to confirm they were realistic, technically valid, and representative of actual risks.

4. Manual Refinement:

- Modified wording to improve clarity and testability of some test steps.
- Simplified ambiguous steps, split complex actions into clearer sub-steps, and aligned expected outcomes to the real behavior of the application.

5. Scenario Consolidation:

- Combined overlapping test cases and removed redundant flows while ensuring coverage completeness.
- Labeled flows explicitly as Basic or Alternate for better traceability in test execution reports.

3.4 Division of Tasks Between AI and Manual Effort

- The development of the scenario testing plan involved a collaborative approach between the AI tool (ChatGPT) and manual effort by me. The following outlines the specific tasks handled by each:
- **Tasks Performed by AI (ChatGPT):**
 - **Initial Scenario Generation:** ChatGPT was used to create the initial drafts of the Basic Flow, Alternate Flows, and Scenario Tables for both the Catalog and Categories features based on the provided prompts. This included generating detailed test scenarios with fields such as Scenario ID, Title, Preconditions, Inputs (Test Data), Test Steps, Expected Result, Actual Result, and Result.
 - **Edge Case Suggestions:** The AI proposed additional edge cases (e.g., invalid inputs, maximum filter selections, special characters in search) to enhance the robustness of the test plans for both Catalog and Categories features.
 - **Bug Report Framework:** ChatGPT generated the initial structure and content for bug reports, including the format with Bug ID, Summary, Steps to Reproduce, Expected Result, Actual Result, Screenshot, Priority, and Severity, based on failed test case results.
- **Tasks Performed Manually:**
 - **Validation and Refinement:** The AI-generated scenarios were manually reviewed and refined to ensure consistency, alignment with functional requirements, and technical validity. This included adjusting ambiguous test steps, splitting complex actions, and clarifying expected outcomes.
 - **Scenario Consolidation:** Overlapping or redundant test cases were manually identified and consolidated to optimize the testing plan while maintaining comprehensive coverage.
 - **Test Execution and Result Logging:** All designed test cases were executed manually, with results (e.g., Actual Result and Result) logged based on the application's behavior during testing.
 - **Bug Reporting Details:** Specific bug reports were manually populated with actual results, priority, and severity assessments, and screenshots were added where applicable, based on the execution outcomes.
 - **Report Compilation and Self-Assessment:** The final report, including group information, scenario definitions, and self-evaluation, was compiled manually, ensuring a professional and traceable document structure.
 - **Integration of New Issues:** Additional bugs (e.g., "Product Names Aligned to the Right," ...) identified during testing were manually incorpo-

rated into the Scenario Table and test cases, with corresponding alternate flows defined.

Chapter 4

Self-Evaluation (Scenario testing)

Criteria	Description	Max Point	Self-assessment
2 Scenario Selection	2 important scenario selected	1.0	1.0
Scenario testing	Correct and complete scenario identification	2.0	2.0
Use of AI Tools	Prompt transparency, critical validation, added value	1.0	1.0
Test Execution	All designed test cases executed, results logged	0.5	0.5
Bug Reporting	Clear and complete bug report(s), if applicable	0.5	0.5
Scenario testing report	Report is clear, traceable, professional, with self-assessment	1.0	1.0